Nov.10, 2026 Semester 1 Class of 2027/28

光剑美高
GUANGHUA ACADEMY

AP Business & Finance

Grade: 11&12
Time: 120 mins

Student Name: _____

Instructions to students:

- No food or drink to be brought into examination room.
- No cell phones/iPods are allowed during the examination at any time. You will be dismissed from the testing site if you are seen with one out. Please do not talk during the examination.
- If you have a problem please raise your hand and wait quietly for a teacher.
- Please do not open the examination booklet until directed to do so.
- Please ensure that you have the correct examination in front of you.
- No pencil cases allowed; bring only the writing materials you need into the examination room.
- Write your name clearly in the space above when directed to do so.
- At the conclusion of your examination please refrain from speaking until you are outside the exam room as there may still be other examinations still in progress.
- Students are reminded that they are not permitted to leave the examination room more than 30 mins early.

Special instructions:

- YOU CAN NOT USE YOUR CALCULATOR FOR THIS SECTION.

学生考试常规要求：

1. 考试前按要求清空教室，考试期间教室里所有与考试关的物品放到储物柜，不得出现在地面、桌椅抽屉内以及桌面上。
2. 考试用品必须在考前准备好，考试时不得借用他人学习用品，若无特殊说明（例如计算器的使用），禁止使用任何电子产品（如电子词典，智能手表，各式通讯设备等等）。
3. 开考前在监考老师检查完考场确定没有问题之后方可入场，按指定座位入座，不得以任何理由交换座位。考试时遵守考场纪律，不得交头接耳或打手势，不准偷看他人试卷，不得出声阅读答案。
4. 如遇试卷分发问题或者试卷字迹模糊可举手询问，但不得要求老师解释试题内容。
5. 考试开始 10 分钟后不得入场，考试剩余 30 分钟以内可以提前交卷。提前交卷者应将试卷反扣留在桌面并及时退出考场，退场时不得带走任何物品。提前交卷者不得在考场附近逗留、讨论、大声喧哗，不得以任何理由在本场考试结束前返回考场。
6. 监考老师示意考试结束之后学生必须立即停笔起立，将试卷反扣留在桌面后及时退出考场。退场过程中严禁喧哗。
7. 旷考者一律作零分论处，并不准补考。因病缺考应出示医院证明，并事先征得任课老师同意，由任课老师酌情处理。
8. 所有考试作弊参与者，除该门成绩作零分计算外，由学校视情节给予记过或者记过以上纪律处分。

诚信考试承诺书

本人郑重承诺如下：

文 **本人珍惜个人荣誉，将诚实守信遵守学校的《学生手册》的规定。**

文 **自觉服从学校教学部门的考试安排，接受监考巡考老师的管理、监督和检查。**

文 **自觉遵守考场纪律，不作弊、不抄袭，不携带违禁电子产品进入考场。**

文 **如有任何疑问和需求，首先需征得监考老师同意。**

本人已认真阅读以上内容，并保证遵守。如有违反，自愿承担相应责任，情节严重愿意承担法律责任。

考生签字： _____

日期： _____

Mid-Term Exam Instructions

- **Time allowed: 120 minutes. Total: 100 points.**
- Answer **all** questions and write in the spaces provided.
- **Section A — Multiple choice (45 points).** Choose the ONE best answer for each question and fill in the circle ◯ before that option. If you fill in more than one circle for a question, that question is not marked.
- **Section B — Short answer (35 points).** Answer about **your own Business Canvas Project**. For each question: **name the concept, define it briefly, then apply it to your own business** using specific evidence from your own customer interviews, market research and product decisions. A conclusion with no explanation earns no credit.
- **Section C — Case analysis (20 points).** The Patagonia case is on the separate sheet provided at the start of this section. Answer both questions below using the case information; the key terms printed under each question are the terms you should use.
- No calculators, phones or other electronic devices.

How to mark your answers in Section A (example)

Fill the circle **completely** with a dark pen or pencil, so that ◯ becomes ●, and leave every circle you are **not** choosing empty.

Example. Which action best protects a small business from running out of raw materials?

Mark	Option
●	**A.** Sign a contract with a second supplier as a backup. — *the answer you chose: the circle is fully filled in*
◯	**B.** Buy all materials from the cheapest supplier. — *not chosen: the circle is left empty*
◯	**C.** Store two years of materials in the stockroom.
◯	**D.** Stop tracking supplier delivery times.

- ✔ **Right:** one circle completely filled in (◯ → ●).
- ✘ **Wrong:** a tick ✓, a slash /, or a half-filled ring ⊘ in the circle; or two or more circles filled in on the same question — that question then scores **0**.
- If you change your mind, cross out the whole line of the old answer and write your new letter in the margin next to the question.

SECTION A — MULTIPLE CHOICE (45 points)

Q1. After two rounds of research a team receives contradictory answers: some students love the app’s design, others say the price is too high, and usage data shows most users drop off after one week. Which is the best professional response?

◯ A. Reject the research as invalid, because a business must never act on contradictory data.

◯ B. Accept the limitation — businesses often cannot accurately predict customer responses and must respond using limited, unclear or contradictory data.

◯ C. Keep only the finding that supports the original idea and ignore the rest.

◯ D. Rerun the same survey with the same sample and questions until the answers agree.

Q2. A shop gives every customer a free repair kit with their purchase. A week later many of those customers return and buy more, saying they felt they owed the shop something. Which principle of influence is at work?

◯ A. Liking — customers are persuaded by people they find similar to them.
◯ B. Consistency — customers want to stay true to their earlier purchase.
◯ C. Consensus — customers follow what most other buyers have done.
◯ D. Reciprocity — people feel obligated to give something back after receiving something of value.

Q3. A business doubles its output. Its revenue rises by 100% while its total cost rises by only 60%, because its fixed costs are now spread across twice as many units. Which statement is correct?

◯ A. Scaling succeeded, because revenue grew faster than cost and each unit now costs less to produce.
◯ B. Scaling failed, because total costs always rise when output doubles.
◯ C. Scaling can only be achieved by building exclusive agreements with suppliers.
◯ D. The business has a cost problem, because costs rose at all.

Q4. A snack brand designs one product line and one message for busy high-school students, instead of trying to appeal to everyone. Which statement best explains the reasoning?

◯ A. Narrow targeting reduces the number of competitors the business has to face.
◯ B. Targeting a specific segment is typically more effective and less costly than appealing to a broad audience.
◯ C. Designing for one segment is always more profitable, because a broad audience can never be profitable.
◯ D. Broad audiences always respond better to marketing than narrow segments do.

Q5. A customer buys a ¥12 coffee on the way to work without comparing options, but carefully compares three laptops before buying one. Which statement is most accurate?

◯ A. The coffee purchase is habitual buying behavior, while the laptop purchase is likely to follow the rational decision-making process.
◯ B. Both purchases are rational, because the customer spends money in both cases.
◯ C. Both purchases are habitual, because the customer is loyal to both brands.
◯ D. The coffee purchase is rational because it meets a need, and the laptop purchase is habitual because laptops are expensive.

Q6. A product launched three months ago has low sales and negative profit, and the company is spending heavily on advertising to build awareness. Which product life cycle stage is it in, and what should the marketing focus be?

◯ A. Decline — sales are falling, so marketing should cut costs or discontinue the product.
◯ B. Growth — sales are rising quickly, so marketing should focus on expanding distribution.
◯ C. Maturity — the market is saturated, so marketing should defend market share with lower prices.
◯ D. Introduction — sales and revenue are typically low, so marketing focuses on tactics that raise brand awareness.

Q7. A student researcher surveys only classmates and finds that 90% would buy an energy drink before class, then charts her monthly website visits. Which statement best identifies both problems?

◯ A. The sample is not representative of the target market, and a pie chart is the wrong tool for trends over time.

◯ B. The sample is too small, and a bar chart should be used to compare discrete data points.

◯ C. There is no problem: classmates are genuine target customers, and any chart can show a trend.

◯ D. The data is qualitative rather than quantitative, so it cannot be charted at all.

Q8. A founder wants to know how many students would pay ¥30 a month for a study-planner app, and why the others would not. Which research design best fits the two questions?

◯ A. An experiment for both questions, because experiments measure behaviour rather than opinions.

◯ B. Observation of existing students only, because what people do matters more than what they say.

◯ C. A focus group for both questions, because group discussion always gives more reliable numbers.

◯ D. A survey for the ‘how many’ question and interviews or a focus group for the ‘why’ question.

Q9. Before buying a restaurant meal, a customer now checks three apps for the lowest price and reads reviews from other customers. Which statement best describes this change?

◯ A. A situational influence, because the customer happened to be standing in the street.

◯ B. Technology and innovation influencing consumer purchasing patterns, on both the buyer and the seller side.

◯ C. A psychological factor, because the customer has learned to prefer reviews.

◯ D. A personal factor, because the customer’s budget has changed.

Q10. A product’s sales have been flat for two years, most customers in the market already own one, and new rivals are cutting prices. The product is in the _____ stage of the product life cycle, and the most appropriate strategy is to:

◯ A. Growth — expand distribution and scale production.

◯ B. Introduction — focus on raising brand awareness.

◯ C. Decline — cut costs or discontinue the product.

◯ D. Maturity — defend market share through differentiation, lower prices, and innovation.

Q11. A bakery fills one order at a time from a handwritten recipe book, while a chain of cafés buys a machine that shapes 2,000 identical buns an hour. Which statement best describes the two production processes?

◯ A. The bakery uses mass production because it produces the same bun every time, and the café chain uses artisan production because the machine is new.

◯ B. Both are mass production, because only artisan production requires skilled labor.

◯ C. The production process is determined by how many products a business sells, not by customer priorities.

◯ D. The bakery uses artisan production (skilled labor, attention to detail, smaller quantities) and the café chain uses mass production (technology, assembly lines, machinery, larger quantities).

Q12. A tea shop posts: “Only 12 tins of this single-origin oolong are left at this price — one tin per customer.” Customers immediately feel urgency to buy. Which principle of influence is the shop using?

◯ A. Scarcity — the rarer an item seems, the more people want it.
◯ B. Consensus — customers follow what other buyers have chosen.
◯ C. Consistency — customers want to stay consistent with their earlier purchases.
◯ D. Authority — customers obey the shop because of its expertise.

Q13. To size the market for running shoes and identify its main competitors, a business reads an industry report and visits competitors’ websites. This is best described as:

◯ A. Primary-source research, because the business collected the information itself.
◯ B. Secondary-source research, because the business is using information that already exists to map the market landscape.
◯ C. A business hypothesis, because the report is an assumption the business will test.
◯ D. Qualitative research, because the report is full of descriptions rather than numbers.

Q14. A business believes students will pay more for a study-planner app with a shared calendar feature than for the same app without it. To test this, it builds both versions, randomly shows each version to real students visiting its site, and records which version produces more paid sign-ups. Which description is most accurate?

◯ A. The business conducted secondary-source research, because it used data that already existed.
◯ B. The business conducted a survey, because it asked a large number of students for their preference.
◯ C. The business tested a hypothesis with an A/B test, a form of primary-source research, and the result is which version produced more sign-ups.
◯ D. The business ran an observation study, because it watched students browse the site without interacting.

Q15. A customer buys the same brand of shampoo every month because it is the cheapest and the store is on the way home. This is best described as:

◯ A. Brand loyalty, because the customer buys the same brand every time.
◯ B. A purchasing pattern based on psychographic characteristics.
◯ C. A habit driven by brand equity, which is why the customer will pay more in future.
◯ D. A repeat purchase rather than brand loyalty, because there is no emotional attachment to the brand.

Q16. A company competes by charging the lowest price in its market. Which supply chain design is most consistent with that strategy?

◯ A. Artisan production using highly skilled labor and exclusive long-term supplier contracts.
◯ B. Mass production with a cost-focused chain that uses cheaper resources and more efficient production processes.
◯ C. Exclusive agreements that prevent suppliers from selling to rivals.
◯ D. A supply chain built on high-quality resources and premium materials.

Q17. After a city bans the sale of energy drinks to minors, a new brand of caffeinated drinks for adults appears, and teenagers switch to it. Which concept best explains the new brand's existence?

◯ A. Market segmentation, because the new brand serves a different demographic.

◯ B. Substitute products — the underlying need persists, so alternatives emerge when one option is restricted.

◯ C. A purchasing pattern, because teenagers now buy drinks more frequently.

◯ D. A barrier to entry, because the ban protects the new brand from competitors.

Q18. A company's logo is recognized and trusted by customers, who choose it over identical products sold by other businesses. The added financial and competitive value created by that recognition and trust is best described as:

◯ A. Product messaging

◯ B. Market saturation

◯ C. Brand equity

◯ D. A value proposition

Q19. A shopper buys a coat partly because her colleagues all own similar coats, and partly because the fitting room is warm and the store is playing music. Which is the best classification?

◯ A. The colleagues are a personal factor; the store atmosphere is a psychological factor.

◯ B. The colleagues are a psychological factor; the store atmosphere is a personal factor.

◯ C. The colleagues are a social and cultural factor; the store atmosphere is a situational influence.

◯ D. Both are personal factors, because they come from the shopper's own life.

Q20. A startup has built a working prototype that customers say they want, but the team cannot find a supplier able to produce the materials at any price, and projected revenue would not cover costs even if it could. Which statement best applies the desirability–feasibility–viability test?

◯ A. The product is viable because customers want it, so the startup should begin mass production immediately.

◯ B. The product fails only the viability test, since the prototype already exists.

◯ C. The product appears desirable, but it fails the feasibility test (it cannot be produced) and the viability test (it cannot profit), so it should not be pursued.

◯ D. The product is feasible because a prototype exists, so the only question left is marketing.

Q21. A bakery's supply chain includes local wheat farms, a mill, the bakery itself, a storage room for finished loaves, and two shops that sell them. Which correctly orders the stages?

◯ A. Raw materials → manufacturing → warehouse → distribution/retail → customer

◯ B. Supplier → distributor → retailer → manufacturing → customer

◯ C. Raw materials → distribution center → manufacturing → warehouse → customer

◯ D. Manufacturing → raw materials → warehouse → customer → retail

Q22. A shopper always buys the unbranded version of ibuprofen because it costs half as much and works the same. This product is best described as:

◯ A. A brand with high brand equity, because the shopper is loyal to it.

◯ B. A differentiated product, because the shopper prefers the plain version.

◯ C. A generic product, sold to customers who prioritise cost savings over brand recognition.

◯ D. An example of a barrier to entry protecting the generic manufacturer.

Q23. Which pairing of customer data tools is correct?

◯ A. Buying customer data from another business is not a legitimate way to collect customer data.

◯ B. Subscriber lists and tracking apps are traditional research tools; surveys and interviews are digital tools.

◯ C. Click-tracking software and social media monitoring are digital tools; surveys and interviews are traditional research tools.

◯ D. Surveys and focus groups are the only sources of quantitative customer data.

Q24. A fitness app wants to market to customers most likely to pay for coaching. It groups users by age, income, and location (demographic) and by whether they value convenience and self-improvement (psychographic), then writes a description of "Lin, 24, a busy graduate student who exercises before work and values quick, guided sessions." Lin's description is best called:

◯ A. A purchasing pattern — the timing, frequency, and quantity of a consumer's purchases.

◯ B. A market segment — the group of users sharing demographic and psychographic traits.

◯ C. A customer profile — a fictional individual who stands for the target segment.

◯ D. Customer data — the raw information collected about users.

Q25. A team sketches a meal-kit box and shows the sketch to 20 classmates, who say the box looks nice. The team then builds a simple version with only five core features and sells it to 50 real customers at full price to see whether they buy. According to the six stages of product development, the sketch belongs to the design stage, and the sellable five-feature version is:

◯ A. A minimum viable product, used in the production stage to test suppliers.

◯ B. A prototype, used in the ideation stage to generate new ideas.

◯ C. A minimum viable product, used in the validation stage to assess product-market fit.

◯ D. A prototype, used in the messaging stage to build brand awareness.

Q26. A clothing brand can buy fabric from three mills: Mill A is the cheapest but sits in a flood-prone region; Mill B is mid-priced with reliable delivery; Mill C is the most expensive and the only Fair Trade–certified supplier. Which statement best reflects how a business should choose a supplier?

◯ A. Always choose Mill C, because quality is the most important factor in every situation.

◯ B. Weigh cost, quality, efficiency, convenience and risk together; choosing on price alone ignores the risk of delays and higher costs.

◯ C. Choose Mill A, because the lowest cost always produces the highest profit.

◯ D. Choose Mill B, because supplier reputation and quality are not supply chain factors.

Q27. A tutoring company has no factory and no warehouse, yet it still has a supply chain. Which combination makes up its service supply chain?

◯ A. Employees, resources, and delivery systems

◯ B. Raw materials, manufacturing, and inventory

◯ C. Suppliers, distributors, and retailers

◯ D. Advertising, market research, and pricing

Q28. A small online shop collects customers' phone numbers for delivery but never tells customers that it also sells those numbers to a marketing partner. Which statement best describes the problem?

◯ A. Effective targeting, because customer data always lowers customer acquisition cost.

◯ B. No problem, because the customers provided the numbers voluntarily.

◯ C. A data breach, because stored customer information has been exposed to outsiders.

◯ D. A privacy violation, because data is collected without the customer's awareness or disclosure; the business must weigh the benefits of customer data against the risks to customers and to its own reputation.

Q29. According to the CED, a business develops its brand identity in order to be consistent with which of the following?

◯ A. Its product's cost structure, so that the brand signals good value.

◯ B. Its vision statement and the product's value proposition, so that the brand appeals to the target customer.

◯ C. Its annual profit target, so that investors recognise the brand.

◯ D. Its competitors' brand identities, so that customers can compare easily.

Q30. A business spends ¥12,000 on marketing, advertising, and sales in a month and acquires 400 new customers. Those customers are expected to spend an average of ¥150 with the business over the next three years. Which statement is correct?

◯ A. CAC is ¥150 per customer and LTV is ¥30, so the business should raise advertising spending.

◯ B. CAC is ¥12,000 and LTV is ¥150, so the business should cut marketing spending.

◯ C. CAC is ¥30 per customer and LTV is ¥150, so building strong relationships is profitable.

◯ D. CAC is ¥30 and LTV is ¥150, but the business should judge CAC alone, since LTV cannot be estimated.

SECTION B — SHORT ANSWER (35 points)

B1. Explain how your business produces its product and delivers it to your customer.

- **(a)** Describe your production process (artisan or mass production), and explain which customer priorities and which core competencies led your team to choose it. (3Pts)
- **(b)** Name your key suppliers and describe each stage of your supply chain, from raw materials (or from the people, resources and delivery systems for a service) to the final customer. (3Pts)
- **(c)** Choose one of the five factors a business weighs when choosing a supplier and explain which one mattered most in your decision. Then identify one risk in your supply chain and explain how your business would manage it. (3Pts)

B2. Explain how your business understands its customers and the data behind them.

- **(a)** Describe the customer data your business collects, name two tools or methods you use to collect it, and explain how demographic and psychographic data each help you decide which customers to serve. (4Pts)
- **(b)** Explain how you used market segmentation to identify your target customers, and describe your customer profile using at least two demographic and two psychographic characteristics. (3Pts)
- **(c)** Identify one privacy, identity-theft or fraud risk your business creates for its customers, and explain one step your business will take to reduce it. (2Pts)

B3. Explain how your customers actually decide to buy, and the tactic you will use to reach them.

- **(a)** When your target customer buys a product like yours, is the decision most likely rational, habitual, or impulsive? Explain your answer using at least three of the following factors: personal, psychological, social and cultural, and situational. (5Pts)
- **(b)** Design ONE sales tactic for your business that is based on at least two of Cialdini's principles of influence. Name each principle you use and explain exactly how your tactic applies it. (4Pts)

B4. Explain the market research your team carried out and how the findings shaped your decisions.

- **(a)** State one business hypothesis your team tested, in the form of a testable assumption about a customer, product, or market. (2Pts)
- **(b)** Describe the primary-source research you conducted: which method(s) you chose and why they match the data needed, the size and the composition of your sample, and two ways you made sure your questions and your sample were unbiased. (4Pts)
- **(c)** Which type of data visualization best presents your most important finding? Explain the insight it gives you about your customer. (2Pts)

SECTION C — CASE ANALYSIS (20 points)

Instructions: Answer **both** questions using the case information on the separate sheet provided and the concepts you have learned in this course. For each answer: **name the concept, define it briefly, then apply it to Patagonia** with specific evidence from the case. The key terms printed under each question are the terms you should use. The case does not illustrate every concept we have studied — answer only what the questions ask.

C1. Patagonia's fleece jacket sells for about three times the price of comparable jackets, and it competes with The North Face, Arc'teryx, Columbia and Decathlon.

Key terms: value creation · value capture · value vs. price · profit = revenue − total cost · differentiation (higher quality, unique features, better customer service, lower price, more effective marketing)

- **(a)** Using **value creation** and **value capture**, explain how Patagonia can charge that price and still make a profit. (5Pts)
- **(b)** Using one or two of the five routes to differentiation, explain how Patagonia achieves competitive advantage in this market. (5Pts)

C2. Patagonia’s purpose statement reads: “We’re in business to save our home planet” and “Earth is now our only shareholder.” In 2020 it laid off 125 employees without severance, then created the One Employee Fund after a public backlash. In 2022 the founding family moved its stock into the Patagonia Purpose Trust and Holdfast Collective.

Key terms: vision vs. mission · core values · business vs. social enterprise vs. nonprofit · ethical dilemma · internal vs. external stakeholder · greatest total benefit / least total harm · corporation: control, liability, funding

- **(a)** Distinguish Patagonia’s **vision** from its **mission**, and decide whether, after the 2022 ownership transfer, Patagonia is a **business**, a **social enterprise**, or a **nonprofit organization**. Justify your answer with the decision tree. (3Pts)
- **(b)** Treat the 2020 layoffs as an **ethical dilemma**: name the internal and the external stakeholders affected, evaluate the benefits and the costs of two possible responses at the time, and recommend a course of action with reasoning. (4Pts)
- **(c)** Explain what Patagonia **gained and what it gave up** by organizing as a **corporation**, and how the 2022 transfer changed **control** without changing the **legal form**. (3Pts)